Duplicate

Starling 2 Max Sys ID Results

This document contains the results of our system identification campaign for the Starling2 Max drone.

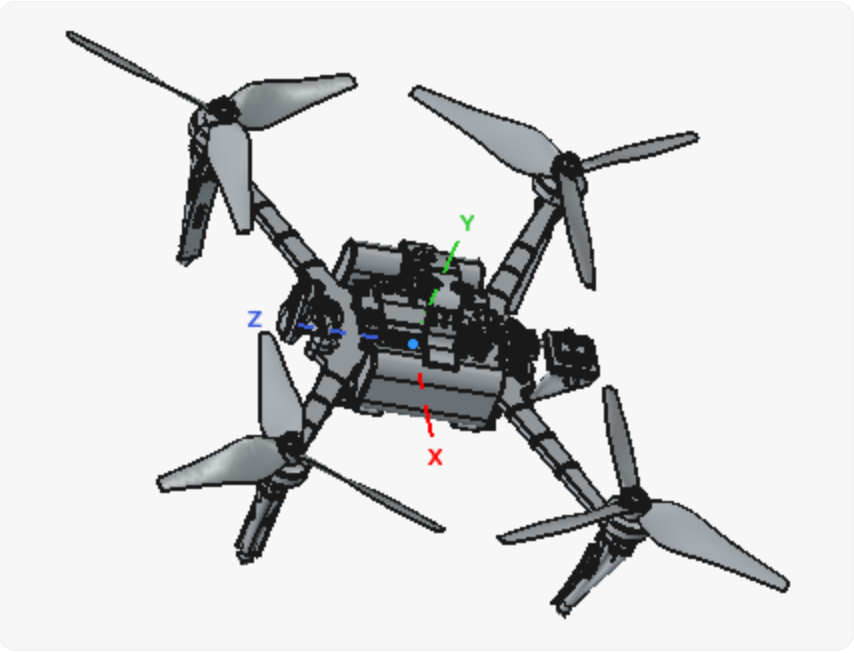
Inertial Parameters

The USD file `starling2max.usd` located below contains all parameters properly set as specified in this section.

Omniverse Navigator
https://airlab-nucleus.andrew.cmu.edu/omni/web3/omniverse://airlab-nu...

Coordinate Frames

The coordinate frame origin and orientation are defined as follows in the ModalAi provided CAD file (see https://modalai.filecloudonline.com/ui/core/index.html#expl-tabl./SHARED/downloads/CAD%20Models)



Where X points left, Z points forward, and Y points up. Please take this unconventional coordinate frame into account (you will likely need to convert the values mentioned here to another frame)

Mass

The total drone mass is **~557g**

Center of Mass

In the CAD coordinate frame, the center of mass is estimated to be at **(0.016mm, 7.41mm, 1.02mm)** (i.e. 0.0016mm left, 7.41mm up, 1.02mm forward). We expect our estimate is accurate up to ~1-2 mm for each axis, so the Y-axis measurement is likely the most important part of this estimate

Inertia Matrix

The inertia matrix **about the center of mass** in g*mm^2 is as follows

Title	X	Y	Z
X	1,857,430.5	-304.3	-2
Y	-304.3	4,649,455.2	-15
Z	-2,655.8	-15,678.1	2,943

Since the off diagonal elements are quite small, we use only the diagonal elements and assume the principal axes are the elementary axes in our USD file.

Propeller Parameters

Rotor Constant

The Thrust vs Prop Angular Velocity equation is $F = (3.964738 \times 10^{-6})\omega^2$, where the rotor constant units are N/(rad/s)^2.

Rolling Moment Coefficient

The Torque vs Prop Angular Velocity equation is $\mathrm{T} = (5.38697 \times 10^{-8})\omega^2$, where the rolling moment coefficient units are Nm/(rad/s)^2

Time Constant

According to step response results, the motor delay constant τ is ~55 ms when increasing throttle and ~85 ms when decreasing throttle. This averages out to **~72 $\pm$ 2.5 ms** under agile throttle changes, as verified from frequency response testing with signal amplitudes swinging between 20% and 50% throttle. **We did not notice significant changes in τ across different battery voltages.**